B2World

Advanced School Management Website & AI-Powered ERP System

Full-Stack Technical Project Assignment

Issued by: B2World
Project Type: Full-Stack Development, ERP, AI Integration, and Database Management
Maximum Duration: 20–25 calendar days
Final Deadline: Within 25 calendar days from the confirmed project start date
Submission: Working application, source code, documentation, testing evidence, and project demonstration

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1. Project Overview

B2World requires the development of a modern, secure, responsive, AI-powered School Management Website and ERP System.

The platform should enable multiple schools to manage their academic, administrative, financial, communication, assessment, and staff-related activities from one centralized system.

The platform must support different user roles, including:

- Platform Super Admin
- School Admin and Principal
- Teachers and Academic Staff
- Non-teaching Staff
- Students
- Parents or Guardians
- HR and Accounts Staff
- Examination Coordinators

Each user must have a role-specific dashboard with access to the information and features permitted for that role.

The system must also include virtual ID cards, online quizzes, examination management, academic performance tracking, events, notifications, analytics, and AI-powered assistance.

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2. Project Objectives

The main objectives are to:

1. Develop a professional public-facing school website.
2. Build a multi-school ERP platform with a Super Admin dashboard.
3. Manage student, staff, teacher, parent, and administrative accounts.
4. Generate digital ID cards for students and staff.
5. Conduct online quizzes, class tests, and other assessments.
6. Provide students and parents with a complete academic performance history.
7. Manage school events, announcements, notices, and notifications.
8. Implement secure fee, attendance, examination, and reporting modules.
9. Integrate useful AI features into the platform.
10. Ensure security, data privacy, usability, and maintainability.

3. Public School Website

Develop a responsive website with the following pages:

- Home
- About the School
- Principal's Message
- Academics and Curriculum
- Classes and Departments
- Admissions and Online Application
- Faculty and Staff Directory
- Events and Activities
- News and Announcements
- Gallery
- Examination Information
- Contact Us
- Careers and Recruitment
- Login and Registration

Requirements

- Responsive design for mobile, tablet, and desktop.
- School branding, logo, contact information, and location.
- Dynamic notices, upcoming events, and admission announcements.
- Contact form with validation and spam protection.
- Search engine-friendly page structure.
- Content management through authorized admin accounts.

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4. Super Admin Dashboard

The Super Admin manages the overall ERP platform and multiple registered schools.

Required features

School and tenant management

- Register and onboard new schools.
- Review and approve school registrations.
- Activate, suspend, or deactivate school accounts.
- Manage school profiles, academic years, and configurations.
- Ensure each school has isolated records and files.

Platform user management

- Manage authorized platform administrators.
- Assign and revoke roles and permissions.
- View account status and login activity.
- Disable compromised or inactive accounts.
- Apply least-privilege access policies.

Subscription management

- Configure subscription plans and trial periods.
- Track subscription status, expiry, and renewals.
- Maintain billing records and subscription reports.
- Clearly distinguish simulated billing from actual payment processing.

Platform analytics

- Total registered schools.
- Active and inactive schools.
- Total students and staff across the platform.
- Usage and feature adoption statistics.
- System errors, service health, and activity summaries.

Security and support

- Audit important administrative actions.
- Review support tickets and platform issues.
- Manage global settings and approved AI configurations.
- Maintain access logs and security alerts.

Super Admin access must not automatically expose every sensitive student, parent, payment, or staff record. Sensitive access must be explicitly authorized and audited.

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5. School Admin and Principal Dashboard

Each school must have its own administrative dashboard.

The school administrator should be able to:

- Manage school details, logo, address, and contact information.
- Configure academic sessions and terms.
- Create classes, sections, subjects, and departments.
- Manage students, teachers, staff, and parent accounts.
- Approve admissions and update student status.
- Assign teachers to classes and subjects.
- Manage timetables and examination schedules.
- Configure fees and payment records.
- Publish notices, events, holidays, and announcements.
- View attendance, academic performance, and financial reports.
- Generate student and staff ID cards.
- Manage user permissions within the school.
- Export authorized reports in PDF or CSV format.

6. Virtual ID Card Generation System

Develop a digital ID card module for students, teachers, non-teaching staff, and other authorized users.

Student ID cards

Each student ID card should display:

- Student photograph.
- Student name.
- Unique student ID or admission number.
- School name and logo.
- Class and section.
- Academic session.
- Blood group, only when voluntarily supplied and appropriately authorized.
- Validity date, if applicable.
- QR code or barcode for identity verification.

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Staff ID cards

Each staff card should display:

- Staff photograph.
- Full name.
- Unique employee ID.
- Designation and department.
- School name and logo.
- Joining date or validity period, where appropriate.
- QR code or barcode for verification.

ID card functionality

- Generate cards automatically from approved profile data.
- Preview cards before issuing them.
- Download or print ID cards in a standard printable format.
- Regenerate cards when authorized profile information changes.
- Deactivate lost, expired, or replaced cards.
- Verify a card using a QR code.
- Display only the minimum information necessary on public verification pages.
- Prevent unauthorized users from creating or editing cards.

Important: QR codes must not contain passwords, private student records, payment details, or sensitive personal data. Use a random, revocable verification token and a protected verification endpoint.

Expected result: An authorized admin can issue an ID card, a student or staff member can view their own card, and a valid QR code can confirm that the card is active.

7. Student Information and Admissions Management

Develop a complete Student Information System.

Features

- Online admission application form.
- Application review and approval workflow.
- Unique admission number generation.
- Student profile and guardian information.
- Class, section, subject, and academic-session assignment.
- Document upload and verification.
- Previous academic record management.
- Student promotion and transfer workflows.
- Student status: active, graduated, transferred, or inactive.
- Search, filtering, and authorized export.

Student records must be protected from access by users belonging to another school or unauthorized role.

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8. Staff, Teacher, and HR Management

The system must maintain staff and teacher records.

Features

- Employee profile and employee ID.
- Department, designation, and joining date.
- Employment status and reporting manager.
- Teacher subject and class assignments.
- Staff attendance and leave requests.
- Leave approval and history.
- Staff notices and internal communications.
- Staff ID card generation.
- Staff training quizzes and assessment results.
- Basic payroll records as an optional enhancement.

Payroll must be access-controlled and must not expose salary information to ordinary staff, students, or parents.

9. Attendance Management

Implement attendance management for both students and staff.

Student attendance

- Daily attendance by class and section.
- Present, absent, late, and approved leave statuses.
- Attendance correction with an audit trail.
- Monthly and term-wise attendance reports.
- Student attendance percentage.
- Configurable attendance thresholds.
- Parent notifications for relevant absences.

Staff attendance

- Daily staff attendance.
- Leave and absence records.
- Monthly summaries.
- Authorized corrections and audit logs.

QR-based attendance may be included as an enhancement. The initial version must not treat a simple scanned QR code as conclusive proof of identity without appropriate validation.

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10. Online Quiz and Assessment System

Build a dedicated assessment module that can be used by students, teachers, administrators, and other authorized staff.

A. Quiz creation

Authorized teachers and assessment coordinators should be able to:

- Create quizzes by subject, class, section, topic, or difficulty.
- Add a quiz title, instructions, duration, total marks, and passing marks.
- Set opening and closing dates.
- Create question banks.
- Add multiple-choice, multiple-select, true/false, short-answer, and descriptive questions.
- Attach images or learning resources where appropriate.
- Configure negative marking, randomization, and attempt limits.
- Preview a quiz before publishing it.

B. Student quiz participation

Students should be able to:

- View assigned and available quizzes.
- Start an eligible quiz.
- See a countdown timer.
- Save answers and resume where the assessment rules permit.
- Submit answers before the deadline.
- Receive a submission confirmation.
- View results and feedback once released by the teacher or system.

The backend must enforce time limits, access windows, attempt limits, and submission rules. These rules must not depend solely on the browser timer.

C. Staff and teacher quizzes

Authorized staff should also be able to take quizzes for:

- Staff training.
- IT and cybersecurity awareness.
- Workplace policies.
- Professional development.
- Subject knowledge and teaching assessments.

Staff quiz results must remain separate from student academic results and should only be visible to authorized staff, trainers, HR, or administrators.

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D. Automated evaluation

- Automatically grade objective questions.
- Calculate marks, percentages, and pass/fail status.
- Provide explanations or feedback where configured.
- Allow teachers to review descriptive answers manually.
- Maintain attempt history and timestamps.
- Record answer changes and submissions where appropriate.

E. Anti-cheating safeguards

- Randomize questions or answer choices where appropriate.
- Enforce attempt limits and deadlines on the server.
- Record suspicious activity for review.
- Avoid claiming that browser monitoring guarantees prevention of cheating.
- Do not use invasive camera or biometric monitoring by default.

F. Quiz reports

Teachers should be able to see:

- Number of students assigned a quiz.
- Number of submissions and pending attempts.
- Average marks and score distribution.
- Question-wise performance.
- Commonly misunderstood topics.
- Individual student results, subject to permissions.

11. Examination and Academic Results Management

Develop a comprehensive academic results system that records performance throughout the school year.

Assessment categories

The system should support configurable assessment types, including:

- Daily or weekly class tests.
- Unit tests.
- Monthly tests.
- Half-yearly or six-monthly examinations.
- Practical examinations.
- Assignments and homework.
- Projects and presentations.
- Online quizzes.
- Annual or final examinations.
- Other school-defined assessments.

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Marks management

Teachers and authorized examination staff should be able to:

- Create examination schedules.
- Enter marks by student, subject, and assessment.
- Set maximum marks and passing marks.
- Record absent, exempt, or pending status.
- Save drafts before publishing results.
- Correct marks through an authorized, auditable process.
- Configure weightage for different assessments.
- Generate subject-wise and overall results.

Student academic dashboard

Every student must have a personal dashboard displaying:

- Class tests and unit test marks.
- Monthly test results.
- Half-yearly examination results.
- Annual or final examination results.
- Online quiz scores.
- Subject-wise marks and percentages.
- Overall term and academic-year performance.
- Grades, where the school's grading scheme supports them.
- Attendance percentage.
- Assignment and homework status.
- Teacher feedback.
- Historical performance comparisons.
- Downloadable report cards.

Students should be able to view their own records, not other students' private marks.

Parent dashboard

Parents or guardians should be able to view the linked child's:

- Published examination results.
- Class test and quiz scores.
- Attendance records.
- Report cards.
- Teacher feedback and academic notices.
- Upcoming examinations and assignment deadlines.

Parents linked to multiple children should be able to switch between their authorized profiles.

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Result calculation

- Allow each school to configure its grading scheme.
- Calculate percentages and weighted totals consistently.
- Clearly show maximum marks and assessment weightage.
- Avoid treating absent or exempt marks as zero unless the school's configured rules require it.
- Keep a record of result publication and authorized corrections.

12. Timetable, Homework, and Learning Management

Timetable

- Class and teacher timetables.
- Subject and classroom allocation.
- Teacher availability checks.
- Examination timetables.
- Timetable updates and notifications.

Homework and assignments

- Teachers can publish homework and assignment instructions.
- Students can submit text or permitted file types.
- Teachers can grade submissions and provide feedback.
- Students can track pending, submitted, and graded assignments.
- Late submissions can be flagged based on configured rules.

Learning resources

- Upload class notes, worksheets, and study materials.
- Organize resources by class, subject, and topic.
- Restrict downloads to authorized users.
- Maintain resource publication dates and versions.

13. Events, Notices, and Latest Updates

Create a centralized events and announcements module.

Event management

Authorized users should be able to publish:

- Annual functions.
- Sports days and competitions.
- Cultural events.
- Parent-teacher meetings.
- Examinations.
- Holidays and school closures.

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- Workshops and seminars.
- Admission deadlines.
- Staff meetings and training.
- Student clubs and extracurricular activities.

Each event should support:

- Event title and description.
- Date, time, and venue or online meeting information.
- Intended audience.
- Registration deadline, if applicable.
- Poster or permitted attachments.
- Organizer details.
- Publication and update timestamps.
- Event status: upcoming, ongoing, completed, or cancelled.

Announcements and notifications

- School-wide notices.
- Class-specific announcements.
- Subject-specific updates.
- Staff-only announcements.
- Parent and student notifications.
- In-app notification center.
- Read/unread status.
- Optional email integration.
- Optional SMS or messaging integration, subject to provider configuration and cost.

The platform should show the latest notices, upcoming events, examinations, deadlines, and urgent announcements on relevant dashboards.

14. Fee and Payment Management

Features

- Configure fee structures by class, term, or academic year.
- Maintain student fee accounts.
- Generate fee invoices and receipts.
- Track paid, pending, overdue, and partially paid amounts.
- Apply authorized discounts or concessions.
- Record refunds and adjustments.
- Generate collection and outstanding-fee reports.
- Notify parents of upcoming or overdue payments.

For the initial demonstration, simulated payment transactions are acceptable if clearly labelled as simulated. Actual payments require a properly configured payment provider, server-side verification, and secure handling of payment events. Never store raw card details.

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15. AI-Powered Features

Implement at least **two working AI features** in the 25-day version. Three are preferred if time permits.

AI feature 1: School Information Assistant

- Answer questions about approved school information.
- Help users find admission procedures, event details, school policies, and timetable information.
- Respect the user's role and permissions.
- Use approved school content and clearly indicate when information is unavailable.
- Avoid inventing school policies, dates, fees, or results.

AI feature 2: Personalized Learning Assistant

- Explain approved learning concepts in age-appropriate language.
- Generate practice questions from approved study materials.
- Provide hints and step-by-step explanations.
- Recommend revision topics based on published assessment performance, when authorized.
- Encourage learning rather than simply giving answers to active examinations.
- Include appropriate safeguards for children and student data.

AI feature 3: AI-Assisted Quiz Generation

- Generate draft questions from teacher-approved notes.
- Suggest difficulty levels, answer options, and explanations.
- Allow teachers to edit and approve every question before publication.
- Clearly identify AI-generated drafts.
- Never publish unreviewed AI-generated assessments automatically.

Optional AI enhancements

- OCR for admission documents, with manual verification.
- Academic trend summaries based on authorized records.
- Suggested revision plans.
- Event announcement drafts.
- Timetable conflict suggestions.
- Early identification of students who may benefit from additional academic support, with human review and no automatic punitive decisions.

AI output must be treated as assistance, not unquestionable fact. Do not expose student records to an AI provider without an approved data-handling approach.

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16. Reports and Analytics

Develop dashboards and reports for authorized users.

School Admin

- Total students and staff.
- Class-wise enrolment.
- Attendance trends.
- Examination performance.
- Fee collection and outstanding balances.
- Upcoming events and examination schedules.
- Quiz participation and score summaries.

Teachers

- Class-wise marks.
- Subject-wise performance.
- Assignment completion.
- Attendance summaries.
- Quiz analysis and topic-level learning gaps.

Students and parents

- Personal academic progress.
- Published test and examination results.
- Attendance trends.
- Upcoming assessments and events.
- Assignment completion status.

Export options

PDF and CSV exports should be available where appropriate, with access checks before generating or downloading sensitive reports.

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17. User Roles and Permissions

Implement role-based access control (RBAC) on the server to ensure that each user can access only the features and records authorized for their role.

1. Super Admin

The Super Admin will have access to platform configuration, school onboarding, platform analytics, and authorized support operations across the platform.

2. School Admin

The School Admin will manage school operations, user accounts, reports, student and staff ID cards, school events, and academic configurations.

3. Principal

The Principal will have access to authorized academic oversight features, school performance monitoring, and school-level reports.

4. Teacher

Teachers will have access to their assigned classes and the ability to manage attendance, quizzes, marks, homework, and class notices.

5. Student

Students will be able to view their own ID cards, quizzes, assignments, attendance records, and published academic results.

6. Parent/Guardian

Parents or guardians will be able to view their linked child's published results, attendance records, fee details, and school notices.

7. HR/Staff Admin

HR and Staff Admin users will manage authorized employee records, leave information, staff quizzes, and staff administration activities.

8. Accounts Staff

Accounts Staff will have access to authorized fee records, invoices, payment receipts, and financial reports.

Security and Access Control Requirements

All role-based permissions must be enforced on the server side, including API endpoints and database operations. Users must not be able to access restricted information by modifying

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record IDs in URLs, query parameters, or API requests. The system must verify each user's role, permissions, and ownership or authorized school-level access before returning or modifying any data.

18. Recommended Technical Stack

The candidate may propose an equivalent stack with justification.

Frontend

- React or Next.js
- TypeScript
- Responsive CSS or a suitable component framework

Backend

- Python
- Django
- Django REST Framework

Database

- PostgreSQL

Optional infrastructure

- Redis and Celery for background tasks
- Cloud object storage for approved documents
- Docker for reproducible development
- Email service for notifications
- An approved LLM or OCR provider for AI features

Other requirements

- Git and GitHub for version control.
- Environment variables for secrets.
- API documentation.
- Automated tests for critical workflows.

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19. Database Design Requirements

Design a relational database with appropriate relationships, constraints, and indexes.

Expected entities include:

- Schools and school settings.
- Users, roles, permissions, and memberships.
- Academic sessions, classes, sections, and subjects.
- Students, parents, guardians, and staff.
- Admissions and student documents.
- Student and staff ID cards.
- Attendance records.
- Timetables and assignments.
- Quiz definitions, questions, choices, attempts, and answers.
- Examination schedules, marks, grading schemes, and report cards.
- Fee structures, invoices, payments, and receipts.
- Events, notices, and notifications.
- AI interaction metadata, where appropriate.
- Audit logs and support tickets.

The design must support multiple schools with strict tenant isolation. Do not rely on frontend filters alone to protect school-specific data.

20. Security, Privacy, and Quality Requirements

The candidate must implement:

- Secure authentication and logout.
- Server-side role and permission checks.
- Password hashing and secure session/token handling.
- Tenant-level authorization on every relevant endpoint.
- Input validation and safe error handling.
- Upload type and size validation.
- Protected student and staff documents.
- Audit trails for sensitive administrative actions.
- Safe handling of marks, fee records, and personal data.
- Configurable data retention and deletion procedures.
- Protection against common web vulnerabilities.
- Backup and recovery considerations.
- No passwords, API keys, or production credentials committed to source control.

Use test or synthetic data for the demonstration. Do not upload real student records or sensitive identity documents to external services without explicit authorization and suitable safeguards.

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21. Testing Requirements

The candidate must demonstrate tests for:

- Login and role permissions.
- School tenant isolation.
- Student and staff ID card creation and verification.
- Quiz creation, timed attempts, submission, and grading.
- Marks entry and result calculation.
- Student access to their own published results.
- Parent access to linked children only.
- Event creation and targeted announcements.
- Fee status and receipt generation.
- AI response behavior and error handling.
- Invalid requests and unauthorized access.

At least the critical workflows must be tested before submission. Provide evidence of the tests performed and known limitations.

22. Deliverables

Submit all of the following:

1. Complete source code repository.
2. Working demo application or local setup instructions.
3. Super Admin dashboard.
4. School Admin and role-specific dashboards.
5. Public website.
6. Student, staff, and parent management modules.
7. Virtual ID card generation and QR verification.
8. Online quiz and assessment module for students and authorized staff.
9. Examination and academic results module.
10. Events, notices, and notification module.
11. Fee management module.
12. At least two working AI features.
13. Database schema, migrations, and ER diagram.
14. API documentation.
15. README with installation and configuration instructions.
16. Automated test code and test results.
17. Sample data or a safe demo account setup.
18. Project report describing architecture, features, and security.
19. Short screen-recorded demonstration.
20. A list of completed features, incomplete features, known bugs, and future improvements.

Do not submit API keys, real passwords, confidential school data, or personal documents.

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23. Maximum 25-Day Development Schedule

Days 1–3: Planning and design

- Requirements clarification.
- UI wireframes and architecture.
- Database design.
- Repository setup and task breakdown.

Days 4–7: Authentication and platform foundation

- Authentication and role permissions.
- Multi-school tenant structure.
- Super Admin dashboard.
- School registration and configuration.

Days 8–11: Core school management

- Student and staff records.
- Admission workflows.
- Student and staff virtual ID cards.
- QR verification.
- Basic attendance and timetable.

Days 12–15: Assessments and academic results

- Quiz creation and question bank.
- Student and staff quiz attempts.
- Automatic grading for objective questions.
- Examination and marks management.
- Student and parent result dashboards.

Days 16–18: Communication and finance

- Events, notices, and notifications.
- Homework and assignments.
- Fee records, invoices, and receipts.
- Essential analytics and reports.

Days 19–21: AI integration

- Implement AI school assistant.
- Implement AI learning assistant or AI quiz generation.
- Add permissions, safeguards, and error handling.
- Integrate AI features into the dashboards.

Days 22–23: Testing and security

- Test critical workflows.

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- Verify tenant isolation and permissions.
- Fix bugs and usability issues.
- Review API validation and file security.

Days 24–25: Final delivery

- Final integration.
- Documentation and database diagrams.
- Demo preparation and screen recording.
- Source code and project report submission.

The schedule is a maximum deadline, not a requirement to wait until day 25 to submit. The candidate should provide regular progress updates and raise blockers early.

24. Evaluation Criteria

Evaluation area	Weight
Core functionality and completeness	15%
Super Admin and multi-school architecture	15%
Virtual ID cards and secure QR verification	10%
Quizzes, examinations, and result tracking	20%
Frontend quality and usability	10%
AI features and responsible implementation	10%
Security, permissions, and data integrity	10%
Testing, documentation, and demonstration	10%
Total	**100%**

25. Acceptance Criteria

The assignment will be considered complete only when the candidate demonstrates the following end-to-end workflows:

1. Super Admin registers and activates a school.
2. School Admin creates a student and staff profile.
3. The system generates a student or staff ID card.
4. A valid QR verification link confirms the card's active status without exposing sensitive data.
5. A teacher creates and publishes a quiz.
6. A student takes the quiz and receives a result according to the configured release rules.
7. Authorized staff can take a staff-training quiz.
8. A teacher records a class test or half-yearly examination result.
9. The student can view their own published marks, subject-wise results, and overall academic progress.
10. A parent can view the linked child's permitted academic records.
11. An authorized user publishes an event or announcement and the intended audience can see it.

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12. An unauthorized user cannot access another school's records or another student's private results.
13. At least two AI features work in the demonstration and handle unavailable information safely.

26. Final Submission Instructions

Submit the repository link, demo link or setup instructions, project report, database ER diagram, API documentation, test evidence, and demonstration video.

Include a final checklist identifying:

- Features completed.
- Features partially completed.
- Features not implemented.
- Known issues.
- Setup requirements.
- Test accounts and their roles.
- AI services used and any limitations.

Deadline: No later than 25 calendar days after the confirmed start date.

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The candidate is expected to build a functional, secure, well-documented application and explain the major technical decisions during the final project review.

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